

Food Delivery Time Prediction – Dual Dataset Study

Clarification to Techanalytic Organisers

We sincerely apologize for any confusion caused during our submission and would like to clarify our dataset usage.

During our experimentation phase, we identified two publicly available datasets with very similar titles related to food delivery time prediction. Although the datasets appear similar at first glance, they differ significantly in feature richness and contextual information.

1. **Food delivery time prediction Case Study Dataset by Bhanuprasad** (Used as Primary Dataset)
2. **Extended Full-Feature Variant** (Used only as Secondary Reference)

Initially, we observed a large performance gap when comparing our results with ($R^2 \approx 0.40$ vs $R^2 > 0.80$). After careful investigation and reviewing author discussions, we discovered that higher scores were achieved using a different dataset variant containing additional contextual features such as weather, traffic, order timing, and city metadata.

To ensure fairness and transparency:

- The dataset provided by the organiser was treated as the **official and primary dataset**.
- All core modelling, evaluation, and conclusions are based on the organiser dataset.
- The extended dataset was used only to understand the performance discrepancy and is clearly marked as secondary.

We sincerely regret any inconvenience caused due to this overlap and have included this clarification to maintain transparency, fairness, and academic integrity in our submission.

Project Overview

This project presents a comparative study of Food Delivery Time Prediction using two different datasets to understand model performance variations and clarify inconsistencies in evaluation metrics.

The goal was to analyze why some public implementations report very high accuracy while baseline datasets produce significantly lower scores.

Project Structure

The project contains two main dataset folders:

1. **Dataset Provided by Organiser (Primary Dataset)**
 2. **Dataset with Full Features (Secondary Dataset)**
-

Dataset 1: Dataset Provided by Organiser (Primary Dataset)

Source:

Bhanuprasad Case Study
<https://www.kaggle.com/datasets/bhanupratapbiswas/food-delivery-time-prediction-case-study>

This dataset was treated as the primary dataset and fully analyzed in our work.

Dataset Columns

- ID
- Delivery_person_ID
- Delivery_person_Age
- Delivery_person_Ratings
- Restaurant_latitude
- Restaurant_longitude
- Delivery_location_latitude
- Delivery_location_longitude
- Type_of_order
- Type_of_vehicle
- Time_taken(min) (*Target Variable*)

Feature Characteristics

- Primarily spatial + basic delivery metadata
- No weather or traffic context
- No temporal delivery pipeline information
- Minimal categorical diversity

This makes it a **low-context tabular dataset**.

Best Model Result (Primary Dataset)

Model: Gradient Boosting Regressor

Approach: Machine Learning without heavy feature engineering

Performance:

- R^2 Score: **0.406**
- RMSE: **±7.23 minutes**

Observation

Despite using multiple Machine Learning and Deep Learning models, the R^2 score remained around **0.30–0.40**, which is relatively low for regression tasks.

This created confusion when comparing with other public implementations showing $R^2 > 0.80$.

Dataset 2: Dataset with Full Features (Secondary Dataset)

Source:

Kaggle notebook dataset variant

<https://www.kaggle.com/code/alperkaraca1/food-delivery-time-prediction/input>

This dataset was used as a secondary reference dataset to understand performance differences.

Additional Columns in This Dataset

In addition to spatial features, this dataset includes rich contextual features:

- Order_Date
- Time_Ordered
- Time_Order_Picked
- Weatherconditions
- Road_traffic_density
- Vehicle_condition
- multiple_deliveries
- Festival

- City

These features introduce:

- Temporal context
- Environmental context
- Operational logistics

Best Model Result (Secondary Dataset)

Model: XGBoost Regressor (Hyperparameter Tuned)

Performance:

- R² Score: **0.8313**
- RMSE: **3.88 minutes**

This represents a major performance improvement compared to the primary dataset.

Comparative Analysis

Factor	Dataset 1 (Organiser)	Dataset 2 (Full Features)
Contextual Features	Low	High
Weather Data	No	Yes
Traffic Data	No	Yes
Temporal Pipeline	No	Yes
Operational Metadata	Limited	Rich
Best R ² Score	0.406	0.83
Prediction Error	Higher	Much Lower

Why the Confusion Happened

Initially, we observed that most models trained on the organiser dataset achieved R² values around **0.35–0.40**, which seemed unusually low compared to Kaggle discussions and public implementations.

We also reviewed several public notebooks based on the same dataset, which similarly reported low R^2 scores.

After reviewing author discussions and alternative notebooks, we discovered that many high-performing implementations used a different dataset variant containing additional contextual features.

This explained the large gap in reported performance.

Final Clarification

- The organiser dataset was correctly analyzed and modeled.
- Lower R^2 values were due to limited feature richness, not modeling errors.
- The secondary dataset was used only for validation and understanding discrepancies.

All primary experimentation and conclusions were derived from:

Folder: Dataset Provided by Organiser

Notes

- Primary analysis is based strictly on organiser-provided data.
- Secondary dataset was used only for validation and clarification purposes.
- All results are reproducible using standard Machine Learning pipelines.